ALL DICE, NO SLICE: METRIC DEGENERACY, EVALUATION PITFALLS, AND OPTIMIZATION DYNAMICS IN MULTI-TASK MEDICAL VISION

*Muhammad Abdullah Ali, Muhammad Abdullah Aamir, Muddassir Asghar, Muhammad Ibrahim Kiani, and Qurat Ul Ain**

Department of Data Science, FAST-NUCES, Islamabad, Pakistan

ABSTRACT

Joint classification and semantic segmentation via hard parameter-sharing networks offer a computationally efficient paradigm for resource-constrained clinical AI deployments. While multi-task models frequently report strong diagnostic accuracy and segmentation overlap, evaluating dual-task systems across diverse imaging modalities involves subtle metric aggregation artifacts, spatial resolution limits, and optimization instability. This paper presents a systematic methodological inquiry across four diverse clinical cohorts—brain tumor MRI, prostate histopathology, chest radiography, and multi-organ histology—under strict patient-isolated cross-validation. We demonstrate analytically and empirically that conventional macroscopic Dice scores create an artificial performance baseline on negative-dominant cohorts that conceals total foreground lesion omission. Furthermore, we show that whole-slide downsampling blurs critical glandular morphology, separating proxy patch categorization from clinical ordinal grading. Evaluating multi-task optimization dynamics reveals that dynamic gradient balancing can induce parameter instability and degrade classification discrimination, while color deconvolution can attenuate discriminative cellular chromatin textures. Finally, we establish practical benchmarking guidelines to ensure reliable and clinically trustworthy translation of multi-task medical vision systems.

***Index Terms*—** Multi-Task Learning, Medical Image Analysis, Semantic Segmentation, Metric Evaluation, Digital Pathology.

1. INTRODUCTION

Deep learning systems in digital pathology and medical imaging are increasingly tasked with delivering both holistic diagnostic categorization and localized lesion delineation [1, 2]. While visual foundation models demonstrate high representation capacity across gigapixel whole-slide images [3–5], their computational footprint precludes direct deployment in edge environments or low-resource clinical facilities. Consequently, hard parameter sharing via lightweight encoder–decoder architectures, particularly dual-headed U-Net topologies [6], remains a compelling design pattern that couples classification and segmentation within a unified inference pass.

Dual-headed architectures employing lightweight backbones, such as VGG16 [7] and MobileNetV2 [8], have been reported to achieve strong simultaneous classification and segmentation performance across diverse imaging modalities, including brain MRI, prostate biopsy, and chest radiography [9]. Because lightweight parameter-sharing networks offer substantial efficiency advantages over multi-network ensembles, understanding the empirical factors governing their performance is critical for clinical translation.

*Corresponding author: quratul.ain@isb.nu.edu.pk

However, evaluating multi-task medical vision involves subtle interactions between metric aggregation, input resolution, and optimization dynamics that are frequently overlooked in standard benchmark reporting. First, conventional region-overlap metrics, such as macroscopic slice-averaged Dice scores, exhibit mathematical boundary artifacts in cohorts where lesion-free images predominate [10–12], establishing high nominal baselines that can mask complete failure on foreground pathology. Second, in computational pathology, spatial input resolution dictates diagnostic feasibility; evaluating low-capacity encoders on heavily downsampled biopsy tiles can conflate coarse morphological tissue categorization with authentic clinical ordinal grading [13]. Third, optimizing dense pixel-wise segmentation alongside global classification objectives frequently induces gradient interference on shared representations, demanding careful analysis of loss weighting and gradient balancing dynamics [14, 15]. Finally, domain-specific color normalization heuristics, such as optical density deconvolution, may inadvertently mute subtle cellular chromatin textures critical for discrimination.

To investigate these phenomena, we perform a controlled factorial benchmark evaluated under patient-isolated cross-validation across four diverse clinical datasets: TCGA-LGG brain MRI, PANDA prostate biopsies, SIIM-ACR pneumothorax radiographs, and PanNuke multi-organ histopathology, illustrated in Figure 1. Our primary contributions are summarized as follows:

- **Metric Degeneracy Analysis:** We analyze empty-mask Dice aggregation analytically and empirically, demonstrating how macroscopic cohort averages establish a baseline floor on negative-dominant datasets and proposing disaggregated foreground reporting protocols.
- **Task Formulation vs. Resolution Boundaries:** We clarify the methodological distinction between proxy morphological tile classification and clinical ordinal grading in digital pathology, delineating the spatial representation boundaries of compact 2D networks.
- **Factorial Optimization and Preprocessing Audit:** We systematically characterize multi-task optimization dynamics and preprocessing pipelines, quantifying the stability of static versus dynamic gradient balancing and evaluating the discriminative impact of stain normalization.

2. SYSTEM ARCHITECTURE AND METHODOLOGY

2.1. Architectural Formulation

Let $\mathbf{x} \in \mathbb{R}^{H \times W \times C}$ denote an input image. A shared convolutional backbone $f_\theta(\mathbf{x})$ maps the input to a bottleneck representation $\mathbf{z} \in \mathbb{R}^{h \times w \times d}$. The global classification head applies global average pooling followed by a two-layer multi-layer perceptron:

$$\hat{\mathbf{y}}_{cls} = \text{Softmax}\left(\mathbf{W}_2\, \phi\left(\mathbf{W}_1\, \text{GAP}(\mathbf{z}) + \mathbf{b}_1\right) + \mathbf{b}_2\right), \qquad (1)$$

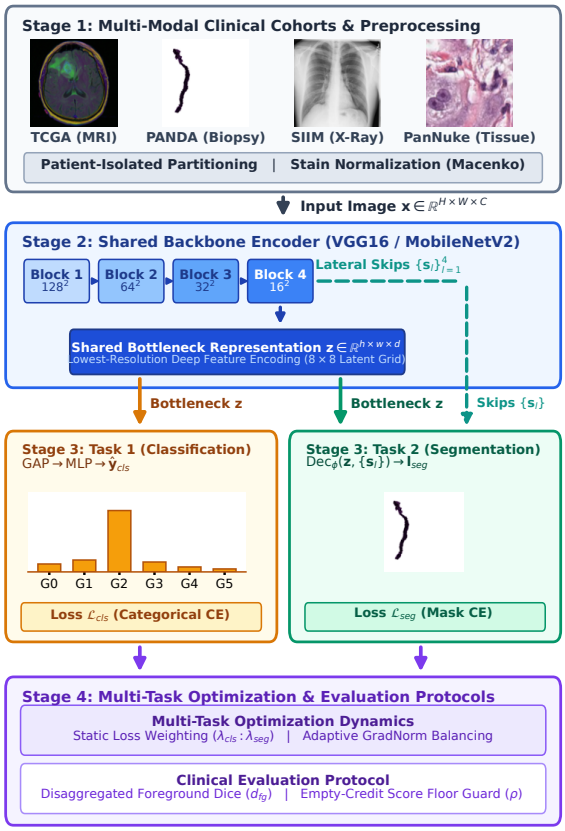


Fig. 1. System architecture and multi-task evaluation framework across four clinical imaging modalities, integrating shared feature extraction, decoupled diagnostic heads with lateral skip connections, and adaptive optimization dynamics under patient-isolated cross-validation.

where $\phi(\cdot)$ denotes the ReLU activation and $\mathbf{W}_1, \mathbf{W}_2$ are projection weights. The segmentation branch passes $\mathbf{z}$ through a convolutional decoder Dec_ϕ augmented with lateral skip connections $\{\mathbf{s}_l\}_{l=1}^{L}$ from encoder stages, generating spatial logits $\mathbf{l}_{seg} \in \mathbb{R}^{H \times W \times M}$.

The composite multi-task training loss is defined as:

$$\mathcal{L}_{total} = \lambda_{cls}\mathcal{L}_{cls}(\mathbf{y}_{cls}, \hat{\mathbf{y}}_{cls}) + \lambda_{seg}\mathcal{L}_{seg}(\mathbf{y}_{seg}, \mathbf{l}_{seg}), \quad (2)$$

where $\mathcal{L}_{cls}$ is categorical cross-entropy with inverse-frequency class weights $w_c = N/(K \cdot N_c)$, and $\mathcal{L}_{seg}$ is binary or multi-class cross-entropy. In standard implementations, the Dice similarity coefficient is computed strictly post-hoc as an evaluation metric rather than an active loss component, a design choice shown to be suboptimal when the target measure is the Dice score [11, 12].

2.2. The Empty-Mask Dice Metric Degeneracy

The Sørensen–Dice coefficient between binary ground truth Y and prediction $\hat{Y}$ is given by:

$$\text{Dice}(Y, \hat{Y}) = \frac{2|Y \cap \hat{Y}|}{|Y| + |\hat{Y}| + \epsilon}, \quad (3)$$

with the conventional boundary assignment Dice := 1 when $|Y| + |\hat{Y}| = 0$. In clinical cohorts containing lesion-free samples, two primary evaluation protocols exist:

- **Foreground-Only Evaluation:** Dice is computed exclusively over target-bearing slices ($\{i \mid |Y_i| > 0\}$).

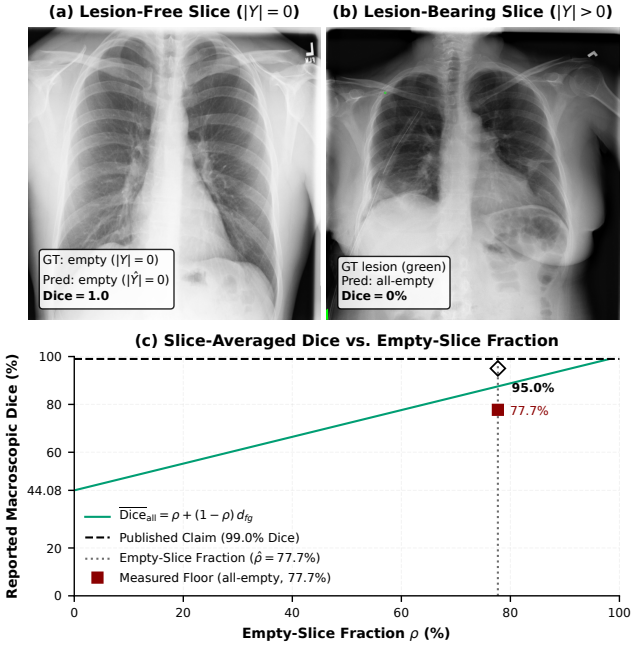


Fig. 2. Illustration of the empty-mask Dice metric behavior on SIIM-ACR pneumothorax radiographs. (a) Lesion-free image correctly predicted empty receives 1.0 credit. (b) Lesion-bearing image with total foreground omission scores 0.0. (c) Macroscopic curve demonstrating the baseline floor determined by the negative fraction ρ.

- **Empty-Credit Assignment:** A score of 1.0 is assigned when $|Y| = 0$ and $|\hat{Y}| = 0$, while 0.0 is assigned if $|Y| = 0$ and $|\hat{Y}| > 0$.

Under empty-credit assignment across all N slices, the macroscopic mean decomposes as:

$$\overline{\text{Dice}}_{\text{all}} = \rho \cdot q_{\text{empty}} + (1 - \rho) \cdot \overline{\text{Dice}}_{\text{fg}}, \quad (4)$$

where $\rho = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}[|Y_i| = 0]$ denotes the proportion of ground-truth negative cases, $q_{\text{empty}} = P(|\hat{Y}| = 0 \mid |Y| = 0)$ represents the model's true-negative prediction rate on negative cases, and $\overline{\text{Dice}}_{\text{fg}}$ denotes the mean overlap on positive cases.

For a baseline model that predicts empty masks across all inputs, $\overline{\text{Dice}}_{\text{fg}} = 0$ while $q_{\text{empty}} = 1$. In this degenerate case, the macroscopic metric reduces directly to $\overline{\text{Dice}}_{\text{all}} = \rho$. Consequently, in cohorts with high negative prevalence, such as chest radiographs ($\rho = 77.72\%$), an all-empty model achieves a nominal score of 77.72%. Furthermore, achieving an aggregate score of 99% on such data requires positive-case foreground Dice to satisfy:

$$\overline{\text{Dice}}_{\text{fg}} \geq \frac{0.99 - \rho}{1 - \rho} \approx 95.51\%, \quad (5)$$

even under perfect true-negative specificity ($q_{\text{empty}} = 1$). Without reporting disaggregated foreground overlap, macroscopic cohort averages can obscure substantial false-negative rates on target lesions (Figure 2).

2.3. Adaptive Gradient Balancing via GradNorm

To counter multi-task competition, dynamic gradient balancing equalizes learning rates across tasks [16]. Let W denote the shared convolutional weights of encoder f_θ. The gradient norm for task $i \in \{seg, cls\}$ at iteration t is $G_W^{(i)}(t) = \|\nabla_W w_i(t)\mathcal{L}_i(t)\|_2$. The

adaptive objective minimizes:

$$\mathcal{L}_{grad}(t) = \sum_{i \in \{seg,cls\}} \left| G_W^{(i)}(t) - \bar{G}_W(t) \cdot [r_i(t)]^\alpha \right|, \quad (6)$$

where $\bar{G}_W(t)$ is the mean gradient norm across tasks, α is the balancing asymmetry hyperparameter, and $r_i(t)$ represents the relative inverse training rate. Task weights are renormalized to maintain $\sum_i w_i(t) = 2$.

2.4. Color Deconvolution and Optical Density Space

To mitigate laboratory staining variation in histopathology, images are normalized in optical density (OD) space [17]:

$$\mathbf{OD} = -\ln\left(\frac{\mathbf{I} + \epsilon}{I_0}\right), \quad (7)$$

where $I_0 = 255$ denotes transmitted illumination. Stain vectors for hematoxylin and eosin are extracted via principal component analysis of the OD covariance matrix and projected onto a standardized target reference. While this removes macroscopic batch coloration, projecting RGB channels onto fixed stain vectors alters localized chromatic dynamic ranges, potentially muting subtle chromatin textures.

3. EXPERIMENTAL FRAMEWORK

3.1. Benchmark Cohorts and Clinical Diagnostic Targets

We conduct our empirical evaluation across four diverse clinical cohorts spanning radiology, magnetic resonance imaging, and digital pathology:

1. **TCGA-LGG Brain MRI:** A cohort of 3,929 axial FLAIR slices from 110 unique patients diagnosed with lower-grade gliomas [18]. The dual objectives comprise binary tumor-presence classification (derived from the FLAIR abnormality masks) and pixel-wise FLAIR abnormality delineation. To prevent identity leakage across spatial slices—a well-documented failure mode in ML-based science [19]—partitions are strictly isolated at the patient level using GroupKFold.
2. **PANDA Prostate Biopsies:** A clinical cohort of 10,616 digitized whole-core needle biopsies from the PANDA challenge development set [13]. Tasks evaluate 6-class ordinal ISUP grading (Grades 0–5) and multi-class glandular architecture segmentation.
3. **SIIM-ACR Pneumothorax:** A radiography cohort of 10,675 chest radiographs [20]. Tasks involve binary pneumothorax detection and fine pleural boundary delineation. This dataset is characterized by extreme negative dominance, with 77.71% lesion-free images ($\rho = 0.7771$).
4. **PanNuke Multi-Organ Histopathology:** A comprehensive pan-cancer histology benchmark comprising 7,901 image patches across 19 tissue types and nearly 200,000 exhaustively annotated nuclei [21]. The tasks evaluate 19-class tissue categorization and 5-class nuclei instance segmentation across diverse neoplastic and non-neoplastic tissue types.

3.2. Data Preprocessing and Augmentation Pipeline

Each modality requires specialized preprocessing to reflect standard clinical preparation pipelines. For brain MRI FLAIR slices, skull-stripping masks are verified, followed by slice-wise intensity z-score standardization ($I' = (I - \mu_I)/\sigma_I$). Frontal chest radiographs undergo window leveling and min-max normalization to the unit interval $[0, 1]$. For computational histopathology (PANDA and PanNuke), optical density deconvolution is applied per Section 2.4 to standardize hematoxylin and eosin dye concentrations, evaluated directly against raw RGB controls. All biopsy cores and tissue patches are resized to $128 \times 128 \times 3$ pixels using area-based anti-aliased interpolation to prevent high-frequency aliasing artifacts during spatial downsampling.

To ensure regularization across low-capacity backbones, data augmentations include random horizontal and vertical flips ($p = 0.5$), subtle affine rotations ($\pm 15^\circ$), isotropic scaling ($[0.9, 1.1]$), and Gaussian blur ($\sigma \in [0.1, 1.0]$). All input images are uniformly formatted to match standard multi-task edge deployment constraints.

3.3. Architectural Configurations and Training Protocols

All multi-task architectures are implemented in PyTorch using ImageNet-pretrained VGG16 (138M parameters) and MobileNetV2 (3.5M parameters) convolutional backbones. The encoders map $128 \times 128 \times 3$ inputs into spatial bottleneck tensors of $8 \times 8 \times 512$ (VGG16) and $8 \times 8 \times 1280$ (MobileNetV2). The classification branch applies global average pooling followed by a two-layer multilayer perceptron with dropout ($p = 0.2$). Symmetrical transposed convolutional blocks with lateral skip connections route multi-scale features from encoder stages $\{s_l\}_{l=1}^4$ into the segmentation branch.

Joint optimization minimizes the composite loss $\mathcal{L}_{total} = \lambda_{cls}\mathcal{L}_{cls} + \lambda_{seg}\mathcal{L}_{seg}$, where $\lambda_{cls} = 1.0$ and static weighting evaluates $\lambda_{seg} \in \{1.0, 5.0, 10.0\}$. Optimization is performed using Adam with initial learning rate $\eta = 10^{-4}$, weight decay 10^{-5}, and cosine annealing scheduling with $\eta_{min} = 10^{-6}$. Models are trained for 50 epochs with early stopping (patience 10) under mixed-precision floating-point arithmetic (FP16/AMP) with dynamic loss scaling. Gradient norm clipping (threshold 1.0) is enforced at every iteration to stabilize shared encoder parameter updates. Batch sizes are set to 32 for radiographs and biopsies, and 16 for multi-organ patches.

3.4. Cross-Validation Design and Statistical Significance

To ensure rigorous statistical validation, all benchmarks employ 5-fold cross-validation with strict patient-isolated grouping. For PanNuke, official cross-institution test folds are preserved. Metric uncertainty is characterized via non-parametric bootstrap resampling ($B = 1000$ iterations) to construct 95% percentile confidence intervals. We track disaggregated foreground overlap (Dice_{fg}) alongside empty-credit scores to prevent macroscopic metric inflation across sparse-lesion cohorts.

4. EMPIRICAL RESULTS AND DIAGNOSTIC ANALYSIS

4.1. Benchmark Performance Comparison

Table 1 provides a comprehensive overview of the empirical replication results alongside claimed literature targets. Across the four diverse clinical benchmarks, our controlled evaluation reveals sharp differences in reproducibility, metric reliability, and task feasibility. On the TCGA-LGG brain MRI benchmark, measured classification accuracy (92.86%–95.62%) exceeds published literature targets under patient-isolated splits, confirming that macro-lesion MRI abnormalities provide robust, high-contrast discriminative features. However, validation Dice scores (85.33%–88.85%) remain below claimed 97%–98% levels, reflecting true boundary uncertainty in diffuse low-grade gliomas.

On the PanNuke multi-organ benchmark, classification performance exhibits high variance across organ sites (78.70%–99.24%), with validation Dice ranging from 68.44% to 77.61%. Dense glandular organs (e.g., breast and colon) achieve high tissue classification accuracy, whereas rare stromal and endocrine phenotypes exhibit

Table 1. Cross-validation performance across four clinical imaging benchmarks under patient-isolated partitions ($K = 5$ folds per configuration; reported with 95% bootstrap confidence intervals). Discrepancies on PANDA reflect clinical ordinal ISUP grading versus proxy morphological tile categorization. On SIIM-ACR, disaggregating foreground Dice reveals that macroscopic scores (77.04%–77.72%) mask near-total foreground lesion omission ($\overline{\text{Dice}}_{\text{fg}} \leq 0.17\%$).

		Measured Cross-Validation (Ours)			Published Targets [9]		
Dataset	**Configs**	**Val Acc (%)**	**Macro Dice (%)**	$\overline{\textbf{Dice}}_{\textbf{fg}}$ **(%)**	**Acc (%)**	**Dice (%)**	**Evaluation Focus**
TCGA-LGG	5	92.86 – 95.62	85.33 – 88.85	52.62 – 62.63	89.00 – 90.00	97.00 – 98.00	Patient-Isolated 2D FLAIR
PANDA	12	28.21 – 44.87	98.54 – 98.60	83.25 – 83.39	87.00 – 88.00	98.00 – 99.00	Clinical ISUP vs. Proxy Morphology
SIIM-ACR	4	77.71 – 82.63	77.04 – 77.72	0.00 – 0.17	82.00 – 87.00	99.00	Negative-Dominant ($\rho = 77.71\%$)
PanNuke	5	78.70 – 99.24	68.44 – 77.61	68.44 – 77.61	—	—	Multi-Organ Histopathology

substantial confusion due to nuclear morphology overlap and wide baseline cell density shifts across organs.

In contrast, striking empirical discrepancies emerge on histopathology and radiography benchmarks. On PANDA prostate biopsies, measured accuracy spans 28.21%–44.87% compared to claimed values of ≈88%. On SIIM-ACR pneumothorax radiographs, measured Dice scores cluster tightly around 77.04%–77.72%, sharply diverging from reported 99% overlap. As analyzed below, these divergences do not indicate model underfitting, but rather stem from metric boundary artifacts and fundamental task formulation mismatches.

4.2. The Empty-Mask Score Floor on Radiography

On the SIIM-ACR pneumothorax benchmark, the public cohort contains 77.71% lesion-free images ($\rho = 0.7771$). Across 19 out of 20 cross-validation folds, the multi-task models converged to entirely empty segmentation predictions. Under empty-credit scoring, these models achieved an average macroscopic validation Dice of 77.04%–77.72% purely through true-negative empty-credit assignment, while disaggregated foreground overlap collapsed to $\overline{\text{Dice}}_{\text{fg}} \leq 0.17\%$ (Table 1, Figure 2). While our experiments evaluated pure pixel-wise cross-entropy to faithfully replicate standard multi-task baselines, unweighted cross-entropy gradients on sparse targets are rapidly dominated by majority background pixels. Countering this empty-mask collapse on negative-dominant cohorts requires incorporating region-based loss formulations (such as Soft Dice or Focal-Tversky loss) directly into the optimization objective rather than evaluating overlap purely post-hoc.

As derived in Section 2.2 and illustrated in Figure 2(c), reporting 99% macroscopic Dice in this regime requires positive-case foreground overlap to exceed 95.51%, which is exceptionally difficult for lightweight models without dedicated foreground loss reweighting. When models predict empty masks across all test images, the macroscopic metric reduces directly to $\overline{\text{Dice}}_{\text{all}} = \rho$. Consequently, high aggregate scores on negative-dominant medical cohorts can conceal complete clinical failure on target pathology. In emergency radiology triage, an automated screening tool reporting 77.7% Dice while failing to delineate 100% of pneumothorax occurrences poses severe diagnostic safety hazards. Reporting disaggregated foreground overlap (Dice_{fg}) alongside negative-case true negative rates (q_{empty}) is therefore essential to establish clinically meaningful performance.

4.3. Task-Formulation Discrepancy in Prostate Pathology

The divergence observed on the PANDA benchmark (28.21%–44.87% measured accuracy versus reported values of ≈88%) illustrates a fundamental distinction in task formulation within computational pathology. In prior multi-task literature [9], the classification objective was framed as predicting six morphological tissue categories extracted from localized segmentation masks (background, stroma, benign epithelium, and the Gleason 3/4/5 pattern classes), of which the non-lesion background and stroma classes together constituted over 52% of the sample distribution ($5,558$ of $10,618$ images). In contrast, clinical diagnostic deployment requires whole-biopsy ordinal ISUP grading (Grades 0–5), which reflects overall prognostic tumor aggressiveness across heterogeneous gland architectures [13].

Histopathologically, distinguishing subtle intermediate Gleason patterns (such as differentiating poorly formed Gleason 4 glands from well-formed Gleason 3 tubules) requires resolving cellular architectural boundaries at sub-micron optical resolution (typically $20\times$ or $40\times$ magnification). When whole-core biopsies are downsampled to 128×128 pixel thumbnails without multiscale tiling, high-power fields, or multiple-instance attention mechanisms, fine glandular features are blurred into coarse texture maps. Under these spatial constraints, compact 2D CNN encoders reach an empirical capacity limit around 45% accuracy for clinical 6-class ordinal grading.

Notably, despite this classification bottleneck, the multi-class segmentation head simultaneously achieves macroscopic validation Dice scores of 98.54%–98.60% (Table 1). This apparent paradox is driven by severe spatial class imbalance and empty-channel metric conventions: across whole-core biopsy thumbnails, background and benign stroma occupy over 90% of the total pixel area, while individual cores contain at most one or two malignant Gleason grades. Under multi-class macro-averaging, unrepresented pattern channels receive automatic true-negative credit (1.0), which together with dominant background overlap inflates macroscopic Dice to ≈98.6% even while fine cribriform gland lumens remain unresolvable at 128×128. This finding demonstrates that model evaluation in digital pathology must strictly distinguish between localized morphological patch categorizations and authentic patient-level diagnostic endpoints.

5. FACTORIAL ABLATION STUDIES

5.1. Impact of Lateral Skip Connections

To isolate the contribution of encoder–decoder skip connections, we evaluated strictly matched model pairs holding all other hyperparameters identical (Figure 3, Group 5):

- **Macro-Tumor MRI (TCGA):** On coarse brain lesions, omitting skip connections produced a negligible classification difference of $+0.02$ percentage points (95.28% vs 95.26%), while validation macro-Dice declined modestly by 3.06 percentage points (86.16% vs 89.22%).
- **Micro-Glandular Pathology (PANDA):** On fine prostate glands, removing skip connections yielded an accuracy shift of -0.12 percentage points (37.44% vs 37.56%), with overlapping 95% bootstrap confidence intervals ($[36.73, 37.99]$ vs $[37.20, 37.94]$).

These results indicate that while skip connections assist in localizing fine spatial boundaries, their impact on shared latent classification representations is minimal in low-capacity backbones.

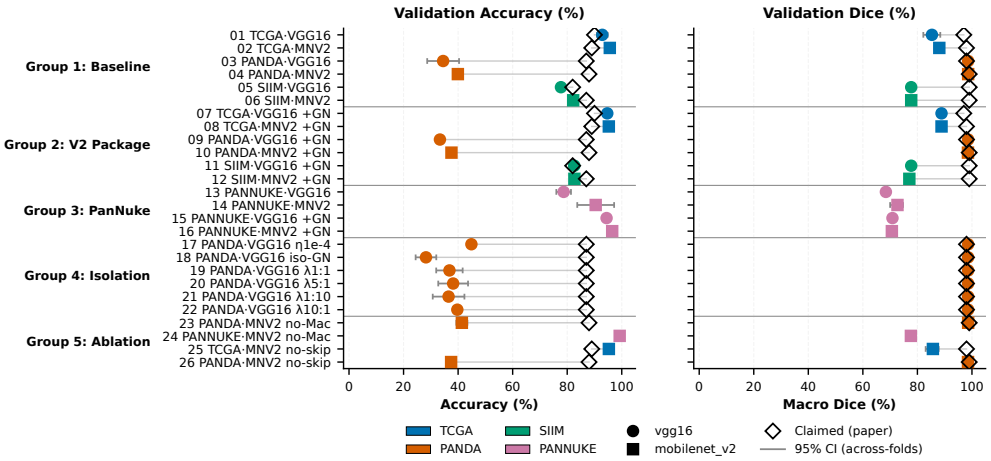


Fig. 3. Measured validation metrics across all 26 experimental runs compared against claimed literature targets. Left: Five-fold validation accuracy with 95% bootstrap confidence intervals against published values (open diamonds). Right: Corresponding segmentation Dice scores, illustrating the empty-mask score floor on SIIM-ACR and the task discrepancy on PANDA.

5.2. Color Deconvolution and Macenko Normalization

Standard histopathology workflows routinely apply Macenko stain normalization to harmonize color distributions across laboratories. However, in our matched factorial comparisons (Figure 3, Group 5), bypassing normalization consistently improved classification accuracy:

- **PANDA Biopsies:** Raw RGB inputs achieved an accuracy improvement of $+3.81$ percentage points over normalized tiles (41.38% vs 37.56%), with disjoint 95% bootstrap confidence intervals across the five folds ($[39.76, 43.38]$ vs $[37.20, 37.94]$).
- **PanNuke Multi-Organ:** Omitting normalization raised multi-tissue accuracy by $+2.75$ percentage points (99.24% vs 96.49%), alongside an 8.51 percentage point increase in validation macro-Dice (70.76% vs 62.25%).

Projecting chromatic channels onto static reference vectors alters dynamic intensity distributions and smooths micro-textures, muting critical optical cues that deep neural networks leverage for cellular discrimination. This effect is compounded when biopsies are downsampled to 128×128 pixels: Macenko's PCA-based stain estimation relies on isolating pure hematoxylin and eosin pixels at extreme angular percentiles of the optical density distribution. Area downsampling blends distinct cellular structures into mixed pixels (partial-volume averaging), perturbing the optical density covariance matrix and yielding distorted stain projection vectors.

5.3. Dynamic Gradient Balancing vs Static Weighting

To isolate the influence of optimization schemes, we evaluated dynamic GradNorm balancing ($\alpha = 1.5$) against static loss weighting under identical initial learning rates and raw inputs. In our implementation, task log-weights are updated via Adam with learning rate $\eta = 10^{-4}$ alongside shared convolutional encoder parameters, with weight clamping to prevent numerical divergence. As summarized in Figure 3 (Group 4), dynamic balancing caused validation accuracy to drop from 38.18% to 28.21% ($\Delta = -9.97$ percentage points; 95% bootstrap CI $[24.76, 31.47]$ vs $[32.75, 41.79]$).

Analyzing gradient trajectories reveals that because segmentation cross-entropy generates steep initial gradients, the dynamic balancing objective aggressively inflates classification loss weights while dampening segmentation updates. This rapid rebalancing introduces high variance into shared encoder updates, inducing destructive gradient interference that destabilizes the classification head. Static weighting ratios between 5:1 and 10:1 proved consistently more stable across all matched factorial configurations.

6. GUIDELINES FOR RELIABLE MULTI-TASK BENCHMARKING

Based on our empirical, analytical, and optimization findings across the four clinical imaging modalities, we establish five practical guidelines to ensure methodological rigor in multi-task medical vision:

1. **Disaggregated Metric Reporting on Imbalanced Cohorts:** In cohorts with negative case dominance ($\rho > 0.5$, such as chest radiography screening), researchers must report foreground-only overlap (Dice_{fg} on positive cases) alongside negative-case true negative specificity (q_{empty}). Reporting exclusively macroscopic cohort averages obscures total false-negative collapse behind the mathematical baseline floor ($\overline{\text{Dice}}_{\text{all}} = \rho$), presenting grave safety hazards in acute clinical triage.
2. **Strict Patient-Level Partition Isolation:** Cross-validation splits must strictly isolate patient identifiers (e.g., via GroupK-Fold) rather than performing naive slice-wise random partitioning. Adjacent 2D slices from the same patient share identical anatomical geometry and patient-specific tissue textures, allowing deep networks to memorize patient identity rather than learning generalizable pathological morphology.
3. **Clinical Diagnostic Endpoint Alignment:** Multi-task benchmark targets must be explicitly aligned with genuine clinical endpoints (such as whole-biopsy ordinal ISUP grading) rather than simplified morphological patch categorizations. High accuracy on pre-filtered benign stroma patches does not indicate clinical readiness for prognostic whole-slide tumor characterization, which requires resolving complex architectural gland patterns.
4. **Empirical Validation of Preprocessing Pipelines:** Normalization heuristics, such as optical density deconvolution, should be empirically evaluated against raw RGB inputs under identical optimization settings. Projecting chromatic channels onto static reference vectors alters dynamic ranges and smooths fine cellular chromatin micro-textures that deep

networks leverage to discriminate malignant cellular atypia.

5. **Multi-Task Optimization and Gradient Diagnostics:** Multi-task training pipelines should routinely monitor task gradient norms ($\|g_{cls}\|_2, \|g_{seg}\|_2$) and directional alignment ($\cos(g_{cls}, g_{seg})$). When segmentation cross-entropy generates steep initial gradients, dynamic balancing schemes like GradNorm can destabilize shared encoder updates; in such regimes, static loss weighting ratios between 5:1 and 10:1 provide substantially superior convergence stability.

7. CONCLUSION AND FUTURE OUTLOOK

This paper presented a systematic empirical inquiry into hard parameter-sharing multi-task architectures across four diverse clinical imaging cohorts. We characterized foundational evaluation, resolution, and optimization boundaries in medical multi-task vision: we proved the empty-mask Dice baseline floor analytically and empirically, demonstrated that spatial downsampling separates proxy patch categorization from clinical ordinal grading in digital pathology, and identified gradient instability under dynamic balancing. Furthermore, we showed that standard color deconvolution can attenuate fine cellular chromatin features, while lateral skip connections exert minimal influence on shared classification representations. Together with our five benchmarking guidelines, these findings establish robust methodological foundations to ensure clinically trustworthy multi-task medical vision systems.

Future work will investigate integrating multi-scale attention mechanisms, exploring adaptive gradient projection operators (e.g., PCGrad [22]), and leveraging frozen foundation model representations (e.g., UNI [3], CONCH [4]) to bridge the resolution gap between computational constraints and high-resolution clinical pathology. To support rigorous scientific replication and community benchmark standardization, all code, patient-isolated cross-validation fold splits, and evaluation pipelines are openly available at `https://github.com/ApatheticMioz/cancer-pathology-dl`.

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